

SAN BERNARDINO, CA, USA

WMO: 722866

Lat: 34.095N

Lon: 117.235W

Elev: 353

StdP: 97.15

Time Zone: -8.00 (NAP)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.4	2.8	-14.6	1.1	12.2	-12.5	1.3	14.7	12.1	13.2	10.5	14.7	0.6	60	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	39.7	20.4	38.0	19.9	36.8	19.7	22.9	35.0	22.0	34.0	21.2	33.2	3.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.3	29.2	17.8	13.3	27.5	17.1	12.7	26.3	69.2	35.3	65.8	33.9	62.6	33.2	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.8	6.2	5.2	DB	-1.2	43.2	1.8	2.3	-2.5	44.8	-3.6	46.2	-4.6	47.5	-5.9	49.2	
			WB	-4.2	24.3	1.2	1.2	-5.1	25.2	-5.8	25.8	-6.4	26.5	-7.3	27.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	12.3	12.9	14.9	18.6	20.1	23.8	27.1	26.9	24.7	21.1	16.3	12.2
	DBStd	6.23	2.96	3.59	3.29	3.17	3.24	3.02	3.08	2.87	3.47	3.30	3.42	3.21
	HDD10.0	42	11	12	2	0	0	0	0	0	0	0	1	16
	HDD18.3	790	188	153	113	34	19	1	0	0	2	11	78	191
	CDD10.0	3432	82	94	156	258	312	415	530	524	442	345	191	84
	CDD18.3	1138	1	2	9	43	73	166	272	266	193	97	18	0
	CDH23.3	14463	50	103	232	702	904	1961	3328	3192	2311	1244	378	57
	CDH26.7	7962	5	18	74	306	422	1086	2014	1914	1344	641	131	7
(14) Wind	WSAvg	1.5	1.2	1.3	1.3	1.5	1.7	1.7	1.8	1.7	1.5	1.3	1.3	1.3
(15) Precipitation	PrecAvg	376	74	70	70	30	9	2	1	1	7	16	47	55
	PrecMax	812	346	318	245	137	69	12	9	10	34	90	215	232
	PrecMin	166	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	175	77	77	56	33	14	3	2	2	10	22	46	56
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.4	29.1	32.4	35.8	37.2	40.3	42.5	41.9	40.7	38.4	32.8	27.7
		MCWB	12.5	14.2	15.6	16.6	17.8	19.9	20.9	21.0	19.5	17.7	14.9	11.6
	2%	DB	24.7	26.5	28.7	32.8	34.1	37.6	39.2	39.0	38.5	35.6	30.2	25.0
		MCWB	11.5	12.8	14.8	16.1	17.8	19.3	21.1	20.5	19.8	16.7	14.0	11.0
	5%	DB	22.5	24.0	26.0	30.1	31.5	35.3	37.6	37.3	36.7	33.1	27.6	22.4
		MCWB	10.8	12.0	13.7	15.6	17.4	19.3	20.4	20.3	19.2	16.0	12.9	10.2
	10%	DB	19.8	21.2	23.0	27.6	28.4	32.9	35.7	35.3	34.1	30.3	25.0	19.9
		MCWB	9.9	11.2	12.6	14.2	16.7	18.9	20.3	20.1	18.8	15.8	12.0	9.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	16.6	17.2	18.8	20.5	22.3	24.1	24.3	22.6	19.9	17.0	14.1
		MCDB	19.6	25.5	26.1	29.7	31.6	34.9	37.5	35.3	35.5	30.3	26.8	18.6
	2%	WB	13.5	14.4	15.9	17.5	19.1	21.3	22.9	23.1	21.5	18.9	15.3	12.8
		MCDB	18.8	20.2	24.9	29.4	30.4	34.2	35.0	34.4	34.4	29.1	24.1	18.6
	5%	WB	12.4	13.2	15.0	16.5	18.2	20.2	21.8	22.1	20.5	18.1	14.5	11.9
		MCDB	18.6	19.8	23.3	27.2	29.2	33.0	33.6	34.1	33.2	28.0	22.9	18.1
	10%	WB	11.4	12.3	14.0	15.6	17.2	19.2	21.0	21.1	19.7	17.3	13.6	11.0
		MCDB	18.0	19.2	20.5	25.6	27.3	31.3	32.8	32.8	31.3	26.3	21.8	17.6
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	12.6	12.7	14.8	14.2	16.8	17.2	17.3	16.7	15.8	15.5	13.7
		MCDBR	18.6	18.8	19.4	20.9	20.4	20.2	18.6	18.5	20.5	21.3	20.1	19.0
		MCWBR	9.1	9.0	8.9	8.8	8.2	7.2	5.8	5.6	7.3	8.6	9.1	9.6
		MCDBR	13.1	14.0	15.5	17.5	17.7	18.9	16.9	17.0	18.0	16.7	15.1	13.7
(40) Clear-Sky Solar Irradiance	5% WB	MCWBR	7.3	7.5	7.9	8.0	7.5	7.8	6.1	6.2	6.7	7.6	7.7	7.7
		taub	0.297	0.307	0.328	0.344	0.357	0.358	0.388	0.371	0.350	0.338	0.304	0.305
		taud	2.539	2.523	2.446	2.407	2.401	2.414	2.396	2.430	2.468	2.472	2.556	2.544
		Ebn at Noon	922	948	948	941	927	921	891	904	912	897	906	890
(44) All-Sky Solar Radiation	Edh at Noon		81	91	107	117	119	117	119	113	104	95	79	77
		RadAvg	3.26	4.17	5.62	6.86	7.59	8.16	7.69	7.25	6.22	4.78	3.63	2.95
(45)	RadStd		0.23	0.39	0.27	0.27	0.35	0.37	0.36	0.22	0.19	0.26	0.16	0.20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		+0.63	N/A	-1.70	+0.72	N/A	N/A	N/A	-112	+223	+112

Nomenclature: See separate page